Mathematics Advanced | MAD 12 | Chapter 1 Series and Sequences

Practice Questions – Applications of AP and GP

Marking Guidelines and Worked Solutions

Notes

- None of these questions were found in the 2020–2025 HSC Mathematics Advanced papers, so all marks and marking criteria are *suggested*, written in NESA style. Total: 34 marks.
- Formulas tagged (Reference Sheet) are quoted from the NESA Reference Sheet:

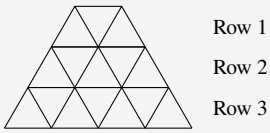
  AP: $T_n = a + (n-1)d, \quad S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$

  GP: $T_n = ar^{n-1}, \quad S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, \ r \neq 1, \quad S = \frac{a}{1-r}, \ |r| < 1$
- Money is rounded to the nearest cent unless the question says otherwise. The **Summary of Final Answers** and the **Mapping Grid** are on the last pages.

Question 1

4 marks

Jay is making a pattern using triangular tiles. The pattern has 3 tiles in the first row, 5 tiles in the second row, and each successive row has 2 more tiles than the previous row.



(i) How many tiles would Jay use in row 20?

(ii) How many tiles would Jay use altogether to make the first 20 rows?

(iii) Jay has only 200 tiles. How many complete rows of the pattern can Jay make?

(i)

1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

The numbers of tiles $3, 5, 7, \ldots$ form an arithmetic sequence with $a = 3$ and $d = 2$.

Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_{20} = 3 + 19 \times 2 = 41$$

$\therefore$ Jay would use 41 tiles in row 20.

(ii)

1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

Using $S_n = \frac{n}{2}(a + l)$ (Reference Sheet) with $a = 3$ and $l = T_{20} = 41$:

$$S_{20} = \frac{20}{2}(3 + 41) = 10 \times 44 = 440$$

$\therefore$ Jay would use 440 tiles altogether.

(iii) 2 marks

Criteria	Marks
• Provides correct solution	2
• Forms a correct inequality or equation for the number of rows, or equivalent merit	1

Sample answer:

Find the number of tiles in the first n rows, using $S_n = \frac{n}{2}[2a + (n-1)d]$ (Reference Sheet):

$$S_n = \frac{n}{2}[2(3) + (n-1)(2)] = \frac{n}{2}(2n+4) = n(n+2)$$

Jay needs $S_n \leq 200$:

$$n^2 + 2n - 200 \leq 0$$

$$\text{Solve } n^2 + 2n - 200 = 0: \quad n = \frac{-2 \pm \sqrt{2^2 + 4 \times 200}}{2} = -1 \pm \sqrt{201}$$

$$n = 13.17\ldots \quad (n > 0)$$

So $n \leq 13.17\ldots$, and n must be a whole number. Check: $S_{13} = 13 \times 15 = 195 \leq 200$, $S_{14} = 14 \times 16 = 224 > 200$.

$\therefore$ Jay can make 13 complete rows.

Question 2 3 marks

The second and fifth terms of a geometric series are 750 and -6 respectively.

(i) Find the common ratio, and the first term of the series.

(ii) What is the limiting sum of the series?

(i) 2 marks

Criteria	Marks
• Provides correct solution	2
• Finds the common ratio, or equivalent merit	1

Sample answer:

Using $T_n = ar^{n-1}$ (Reference Sheet):

$$T_2 = ar = 750 \quad \cdots (1) \qquad T_5 = ar^4 = -6 \quad \cdots (2)$$

Divide (2) by (1):

$$r^3 = \frac{-6}{750} = -\frac{1}{125} \quad \Longrightarrow \quad r = -\frac{1}{5}$$

Substitute into (1): $a \times \left(-\frac{1}{5}\right) = 750$, so $a = -3750$.

$\therefore$ The common ratio is $-\frac{1}{5}$ and the first term is -3750.

(ii)

1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

Since $|r| = \frac{1}{5} < 1$, the limiting sum exists. Using $S = \frac{a}{1-r}$ (Reference Sheet):

$$S = \frac{-3750}{1-\left(-\frac{1}{5}\right)} = -3750 \div \frac{6}{5} = -3750 \times \frac{5}{6} = -3125$$

Question 3

1 mark

Evaluate $\sum_{n=2}^{4} n^2$

Criteria	Marks
• Provides correct answer	1

Sample answer:

Substitute $n = 2, 3, 4$ and add:

$$\sum_{n=2}^{4} n^2 = 2^2 + 3^2 + 4^2 = 4 + 9 + 16 = 29$$

Question 4

3 marks

The seats in a theatre are numbered in numerical order from the first row to the last row. The first row has 14 seats. Each successive row has 4 more seats than the previous one.

(a) How many seats are there in the fourth row?

(b) The theatre has a maximum capacity of 1320 seats, what is the greatest number of rows of seats in the theatre?

(a)

1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

The numbers of seats in the rows form an arithmetic sequence with $a = 14$ and $d = 4$.

Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_4 = 14 + 3 \times 4 = 26$$

$\therefore$ There are 26 seats in the fourth row.

(b) **2 marks**

Criteria	Marks
• Provides correct solution	2
• Forms a correct inequality or equation for the number of rows, or equivalent merit	1

Sample answer:

The number of seats in the first n rows is, using $S_n = \frac{n}{2}[2a + (n-1)d]$ (Reference Sheet):

$$S_n = \frac{n}{2}[2(14) + (n-1)(4)] = \frac{n}{2}(4n + 24) = 2n^2 + 12n$$

The theatre holds at most 1320 seats, so $S_n \leq 1320$:

$$2n^2 + 12n \leq 1320$$
$$n^2 + 6n - 660 \leq 0$$

$$\text{Solve } n^2 + 6n - 660 = 0: \quad n = \frac{-6 \pm \sqrt{6^2 + 4 \times 660}}{2} = \frac{-6 \pm \sqrt{2676}}{2}$$
$$n = 22.86\ldots \quad (n > 0)$$

So $n \leq 22.86\ldots$, and n must be a whole number. Check: $S_{22} = 2(22)^2 + 12(22) = 1232 \leq 1320$, $S_{23} = 2(23)^2 + 12(23) = 1334 > 1320$.

$\therefore$ The greatest number of rows is 22.

Question 5

2 marks

A staircase is designed so that the height of each step increases by a constant amount. The first step is 15 cm high, and the fifth step is 25 cm high.

(a) Find the common difference in the step heights.

(b) Calculate the height of the 10th step.

(a) **1 mark**

Criteria	Marks
• Provides correct answer	1

Sample answer:

The step heights form an arithmetic sequence with $a = 15$. Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_5 = a + 4d = 25 \quad \Rightarrow \quad 15 + 4d = 25 \quad \Rightarrow \quad 4d = 10 \quad \Rightarrow \quad d = 2.5$$

$\therefore$ The common difference is 2.5 cm.

(b) **1 mark**

Criteria	Marks
• Provides correct answer	1

Sample answer:

Using $T_n = a + (n-1)d$ (Reference Sheet) with $a = 15$ and $d = 2.5$:

$$T_{10} = a + 9d = 15 + 9 \times 2.5 = 37.5$$

$\therefore$ The 10th step is 37.5 cm high.

Question 6

3 marks

A biologist is growing a cell culture. The culture doubles in size **every hour**. If the initial population is 500 cells:

(a) Find the population after 5 hours.

(b) After how many hours will the population first exceed 40 000 cells?

(a)

1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

The populations 500, 1000, 2000, … (at the start, after 1 hour, after 2 hours, …) form a geometric sequence with $a = 500$ and $r = 2$.

Using $T_n = ar^{n-1}$ (Reference Sheet), the population after n hours is the $(n+1)$th term: $T_{n+1} = 500 \times 2^n$.

$$\text{After 5 hours:} \quad 500 \times 2^5 = 500 \times 32 = 16\,000$$

$\therefore$ The population after 5 hours is 16 000 cells.

(b)

2 marks

Criteria	Marks
• Provides correct solution	2
• Forms the inequality $500 \times 2^n > 40\,000$, or equivalent merit	1

Sample answer:

The population exceeds 40 000 when

$$\begin{aligned} 500 \times 2^n &> 40\,000 \\ 2^n &> 80 \\ n &> \log_2 80 = \frac{\ln 80}{\ln 2} = 6.32\ldots \qquad \text{(change of base (Reference Sheet))} \end{aligned}$$

Since the population doubles at the end of each hour, n is a whole number, so $n = 7$.

Check: after 6 hours, $500 \times 2^6 = 32\,000 < 40\,000$; after 7 hours, $500 \times 2^7 = 64\,000 > 40\,000$.

$\therefore$ The population first exceeds 40 000 cells after 7 hours.

Question 7

3 marks

A new car is purchased for \$35 000. It depreciates in value by 12% each year.

(a) Find the value of the car after 6 years.

(b) What is the total percentage loss in value after 6 years?

(a)

1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

Each year the value is multiplied by $1-0.12 = 0.88$, so the values form a geometric sequence with ratio $r = 0.88$. After n years the value is $35\,000 \times 0.88^n$.

After 6 years: $V = 35\,000 \times 0.88^6 = 16\,254.143\ldots$

$\therefore$ The value of the car after 6 years is \$16 254.14.

(b)

2 marks

Criteria	Marks
• Provides correct answer	2
• Finds the loss in value, or equivalent merit	1

Sample answer:

First find the loss in value, then express it as a percentage of the original value:

$$\text{Loss in value} = 35\,000 - 16\,254.14 = \$18\,745.86$$

$$\text{Percentage loss} = \frac{18\,745.86}{35\,000} \times 100\% = 53.56\ldots\%$$

(Equivalently, the percentage loss is $\left(1 - 0.88^6\right) \times 100\% = 53.56\ldots\%$.)

$\therefore$ The total loss in value after 6 years is about 53.6%.

Question 8

5 marks

Helga is training for a marathon. Her training includes a run every Sunday, starting with a run of 3 km on the first Sunday. Each Sunday she increases the length of her run from the previous Sunday by 2 km.

(i) Show that on the 5th Sunday of training, she runs 11 km.

(ii) Find a simplified expression for length of her training run on the nth Sunday.

(iii) On the nth Sunday, Helga runs 43 km. Find the value of n.

(iv) Find the total distance she runs on Sundays, in 15 weeks of training.

(i)

1 mark

Criteria	Marks
• Provides correct solution	1

Sample answer:

The lengths of the runs form an arithmetic sequence with $a = 3$ and $d = 2$.

Using $T_n = a + (n-1)d$ (Reference Sheet): $T_5 = 3 + 4 \times 2 = 11$

$\therefore$ On the 5th Sunday she runs 11 km, as required.

(ii) 1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

Using $T_n = a + (n-1)d$ (Reference Sheet) with $a = 3$ and $d = 2$:

$$T_n = 3 + (n-1) \times 2 = 3 + 2n - 2 = 2n + 1$$

$\therefore$ On the nth Sunday she runs $(2n + 1)$ km.

(iii) 1 mark

Criteria	Marks
• Provides correct answer	1

Sample answer:

Using the expression from part (ii):

$$2n + 1 = 43 \quad \Longrightarrow \quad 2n = 42 \quad \Longrightarrow \quad n = 21$$

(iv) 2 marks

Criteria	Marks
• Provides correct answer	2
• Substitutes correctly into a formula for S_n, or equivalent merit	1

Sample answer:

In 15 weeks there are 15 Sundays. Using $S_n = \frac{n}{2}[2a + (n-1)d]$ (Reference Sheet):

$$S_{15} = \frac{15}{2}[2(3) + 14(2)] = \frac{15}{2} \times 34 = 255$$

$\therefore$ She runs a total of 255 km on Sundays.

Question 9 4 marks

A wildlife camera records video clips over consecutive nights. The number of clips recorded each night forms an arithmetic sequence. Over the first 10 nights the camera records a total of 730 clips. On the tenth night it records 100 clips.

(a) Show that the number of clips recorded on the first night is 46.

(b) Find the common difference of the sequence.

(c) Assuming the same pattern continues, find the first night by which the total number of clips recorded exceeds 1200.

(a) 1 mark

Criteria	Marks
• Provides correct solution	1

Sample answer:

Using $S_n = \frac{n}{2}(a + l)$ (Reference Sheet) with $n = 10$, $S_{10} = 730$ and $l = T_{10} = 100$:

$$730 = \frac{10}{2}(a + 100) \quad \Longrightarrow \quad 146 = a + 100 \quad \Longrightarrow \quad a = 46$$

$\therefore$ The camera records 46 clips on the first night, as required.

(b) **1 mark**

Criteria	Marks
• Provides correct answer	1

Sample answer:

Using $T_n = a + (n-1)d$ (Reference Sheet):

$$T_{10} = a + 9d = 100 \quad \Rightarrow \quad 46 + 9d = 100 \quad \Rightarrow \quad 9d = 54 \quad \Rightarrow \quad d = 6$$

(c) **2 marks**

Criteria	Marks
• Provides correct solution	2
• Forms a correct inequality or equation for the number of nights, or equivalent merit	1

Sample answer:

The total number of clips in the first n nights is

$$S_n = \frac{n}{2}\left[2(46) + (n-1)(6)\right] = \frac{n}{2}(6n + 86) = 3n^2 + 43n$$

The total exceeds 1200 when $S_n > 1200$:

$$3n^2 + 43n - 1200 > 0$$

$$\text{Solve } 3n^2 + 43n - 1200 = 0: \quad n = \frac{-43 \pm \sqrt{43^2 + 4 \times 3 \times 1200}}{2 \times 3} = \frac{-43 \pm \sqrt{16\,249}}{6}$$

$$n = 14.07\ldots \quad (n > 0)$$

So $n > 14.07\ldots$, and the first whole number is $n = 15$.

Check: $S_{14} = 3(14)^2 + 43(14) = 1190 \leq 1200$, $S_{15} = 3(15)^2 + 43(15) = 1320 > 1200$.

$\therefore$ The total first exceeds 1200 clips on night 15.

Question 10

6 marks

A charity records the donations received each week for the first four weeks of a fundraising drive, as shown. The director notices that the weekly donations follow a geometric sequence and assumes the pattern will continue.



(a) Find the common ratio.

(b) How much would she predict to be received in Week 8 (to the nearest dollar)?

(c) What total amount would be raised in the first thirteen weeks?

(d) She claims the drive will never raise more than \$8100 in total. Explain how she can make this prediction and why she decides to launch a new drive after thirteen weeks.

(a) **1 mark**

Criteria	Marks
• Provides correct answer	1

Sample answer:

Divide a week's donation by the previous week's donation:

$$r = \frac{1800}{2700} = \frac{2}{3} \qquad \left(\text{check: } \frac{1200}{1800} = \frac{800}{1200} = \frac{2}{3}\right)$$

(b) **1 mark**

Criteria	Marks
• Provides correct answer	1

Sample answer:

Using $T_n = ar^{n-1}$ (Reference Sheet) with $a = 2700$ and $r = \frac{2}{3}$:

$$T_8 = 2700 \times \left(\frac{2}{3}\right)^7 = 158.02\ldots$$

$\therefore$ She would predict \$158 in Week 8.

(c) **2 marks**

Criteria	Marks
• Provides correct answer	2
• Substitutes correctly into the formula for S_n, or equivalent merit	1

Sample answer:

Using $S_n = \dfrac{a\,(1 - r^n)}{1 - r}$ (Reference Sheet):

$$S_{13} = \frac{2700\left(1 - \left(\frac{2}{3}\right)^{13}\right)}{1 - \frac{2}{3}} = 8100\left(1 - \left(\frac{2}{3}\right)^{13}\right) = 8058.38\ldots$$

$\therefore$ The total raised in the first thirteen weeks would be \$8058.38.

(d) **2 marks**

Criteria	Marks
• Provides a correct explanation of both the claim and the decision	2
• Finds the limiting sum of \$8100, or equivalent merit	1

Sample answer:

Since $|r| = \frac{2}{3} < 1$, the series of donations has a limiting sum. Using $S = \dfrac{a}{1 - r}$ (Reference Sheet):

$$S = \frac{2700}{1 - \frac{2}{3}} = 2700 \times 3 = 8100$$

The total raised gets closer and closer to \$8100 as the weeks go on, but never exceeds it. So she can predict that the drive will never raise more than \$8100.

After thirteen weeks the total is already \$8058.38, which is over 99% of \$8100. All the remaining weeks together could add at most 8100 − 8058.38 = \$41.62, and the weekly donations are now very small (Week 13: $2700 \times \left(\frac{2}{3}\right)^{12} \approx \$20.81$). So it is worth launching a new drive rather than continuing this one.

Summary of Final Answers

Question	Final answer	Marks
1 (i)	41 tiles in row 20	1
1 (ii)	440 tiles in the first 20 rows	1
1 (iii)	13 complete rows $(S_n = n(n+2);\ S_{13} = 195, S_{14} = 224)$	2
2 (i)	$r = -\frac{1}{5},\ a = -3750$	2
2 (ii)	Limiting sum $= -3125$	1
3	$\sum_{n=2}^{4} n^2 = 4 + 9 + 16 = 29$	1
4 (a)	26 seats in the fourth row	1
4 (b)	22 rows $(S_n = 2n^2 + 12n;\ S_{22} = 1232, S_{23} = 1334)$	2
5 (a)	$d = 2.5$ cm	1
5 (b)	The 10th step is 37.5 cm high	1
6 (a)	16 000 cells	1
6 (b)	After 7 hours $(500 \times 2^n > 40\,000$ gives $n > 6.32\ldots)$	2
7 (a)	\$16 254.14	1
7 (b)	About 53.6% (loss of \$18 745.86)	2
8 (i)	$T_5 = 3 + 4 \times 2 = 11$ km (shown)	1
8 (ii)	$T_n = 2n + 1$ km	1
8 (iii)	$n = 21$	1
8 (iv)	255 km	2
9 (a)	$730 = 5(a + 100)$ gives $a = 46$ (shown)	1
9 (b)	$d = 6$	1
9 (c)	Night 15 $(S_n = 3n^2 + 43n;\ S_{14} = 1190, S_{15} = 1320)$	2
10 (a)	$r = \frac{2}{3}$	1
10 (b)	\$158	1
10 (c)	\$8058.38	2
10 (d)	Limiting sum $S = \frac{2700}{1 - \frac{2}{3}} = \$8100$, so the total never exceeds \$8100. After 13 weeks the total is already over 99% of \$8100 and weekly donations are tiny, so a new drive is worthwhile.	2
	Total	**34**

Mapping Grid

Question	Marks	Content	Syllabus outcomes
1 (i)	1	MA-M1 Modelling Financial Situations (arithmetic sequences)	MA12-4
1 (ii)	1	MA-M1 Modelling Financial Situations (arithmetic series)	MA12-4
1 (iii)	2	MA-M1 Modelling Financial Situations (arithmetic series)	MA12-4
2 (i)	2	MA-M1 Modelling Financial Situations (geometric sequences)	MA12-4
2 (ii)	1	MA-M1 Modelling Financial Situations (limiting sum)	MA12-4
3	1	MA-M1 Modelling Financial Situations (sigma notation)	MA12-4
4 (a)	1	MA-M1 Modelling Financial Situations (arithmetic sequences)	MA12-4
4 (b)	2	MA-M1 Modelling Financial Situations (arithmetic series)	MA12-4
5 (a), (b)	2	MA-M1 Modelling Financial Situations (arithmetic sequences)	MA12-4
6 (a)	1	MA-M1 Modelling Financial Situations (geometric sequences)	MA12-4
6 (b)	2	MA-M1 Modelling Financial Situations (geometric sequences, logarithms)	MA12-4
7 (a), (b)	3	MA-M1 Modelling Financial Situations (depreciation as a geometric sequence)	MA12-2, MA12-4
8 (i)–(iii)	3	MA-M1 Modelling Financial Situations (arithmetic sequences)	MA12-4
8 (iv)	2	MA-M1 Modelling Financial Situations (arithmetic series)	MA12-4
9 (a), (b)	2	MA-M1 Modelling Financial Situations (arithmetic sequences and series)	MA12-4
9 (c)	2	MA-M1 Modelling Financial Situations (arithmetic series)	MA12-4
10 (a)–(c)	4	MA-M1 Modelling Financial Situations (geometric sequences and series)	MA12-4
10 (d)	2	MA-M1 Modelling Financial Situations (limiting sum)	MA12-4, MA12-10
	34		

All marks are suggested. MA12-2: models and solves problems and makes informed decisions about financial situations using mathematical reasoning and techniques. MA12-4: applies the concepts and techniques of arithmetic and geometric sequences and series in the solution of problems. MA12-10: constructs arguments to prove and justify results and provides reasoning to support conclusions which are appropriate to the context.